Experiment 4: Total Sales and Percentage Increase

Aim

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy arrays and arithmetic operations.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array named sales_data containing sales for four quarters.
4. Calculate the total sales using np.sum().
5. Calculate the percentage increase from the first quarter to the fourth quarter.
6. Display the total sales and percentage increase.
7. Stop the program.

Input

Quarter 1 sales = 10000

Quarter 2 sales = 12000

Quarter 3 sales = 15000

Quarter 4 sales = 18000

Code

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0]) /       sales_data[0]) * 100

print("Total sales for the year:", total_sales)

print("Percentage increase:", percentage_increase, "%")
```

Output

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 5.py ==========
Enter sales for Quarter 1: 10000
Enter sales for Quarter 2: 12000
Enter sales for Quarter 3: 15000
Enter sales for Quarter 4: 18000
Total sales for the year: 55000.0
Percentage increase: 80.0 %
```

Result

Thus, the total sales for the year were calculated as **55000**, and the percentage increase in sales from the first quarter to the fourth quarter was **80%**.